

CADMIUM AND MANGANESE SORPTION BY SOIL MACROPORE LININGS AND FILLINGS¹

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We collected materials partially filling or coating the walls of soil macropores to compare their sorptivity for cadmium (Cd) and manganese (Mn) with that in homogenized soil samples from the same soil profile. Organic-rich linings and fillings in root-derived soil macropores exhibited higher sorptivities for Cd and Mn than homogenized soil, particularly at lower solution pH values (pH 4 to 5). Due to the higher density of pH-dependent sorption sites on organic matter, sorption on root-derived macropore materials increased more rapidly as pH increased than did sorption on inorganic macropore materials. Inorganic linings and fillings in fracture-derived macropores exhibited sorptivities for Cd and Mn that were very similar to homogenized soil sorptivities. For both organic and inorganic macropore materials, solution pH was generally far more important in determining sorptivity than the nature of the macropore material. Overall, the results of the study suggest that the migration of Cd and Mn will likely be more highly retarded in root-derived macropores than in other types of macropores. The absence of significant differences in sorptivity between inorganic macropore materials and homogenized soil suggests that factors (e.g., hydrology) other than sorptivity will be more important in determining whether retardation of Cd and Mn is greater or less in the whole soil matrix than that in non-root-derived macropores.

Sorption characteristics of homogenized whole soils have been used to estimate retardation of contaminant transport in soil (e.g., Baes

and Sharp 1983). Nonetheless, there is a growing recognition that movement of contaminants through preferred channels, or macropores, in field soils may bypass much of the soil matrix and lead to faster contaminant transport to groundwater and streams than predicted from whole soil properties (Shuford et al. 1977; Thomas and Phillips 1979). Water moving relatively slowly through the microporous matrix of soil will provide the maximum opportunity for sorption of soluble contaminants. In contrast, water moving rapidly through macropores (such as burrows, fossil root channels, and desiccation fractures) provides less opportunity for sorption due to both the shorter contact time and the less intimate contact with sorbing surfaces. It is reasonable to hypothesize that materials that partially fill or coat the walls of macropores may have different sorption characteristics than the whole soil. For example, where a macropore has been derived from root decay, the presence of residual organic matter may be expected to increase locally the sorption of certain contaminants. Where a macropore has resulted from desiccation or structural development, the quantities and types of clay and iron (Fe), manganese (Mn), and aluminum (Al) hydrous oxides in the macropore milieu may vary from the whole soil such that sorption is locally increased or decreased. Whether sorption is actually increased or decreased by passage of water through a macropore will also depend on the kinetics of the sorption reaction and on the intimacy of contact between the dissolved contaminant and macropore materials.

The objective of the research described here was to compare the trace element sorptivity of materials partially filling or coating the walls of soil macropores with homogenized whole soil samples from the same soil profile. Both the soil profiles chosen for study are located on the Oak Ridge Reservation (ORR) and are being instrumented for detailed studies and modeling subsurface transport of trace elements. Cadmium (Cd) and manganese were chosen for study because they were expected to exhibit relatively low retardation in subsoils of the ORR and because radioisotopes of these elements offer

¹ Research sponsored by the Office of Health and Environmental Research, U.S. Dept. of Energy, under Contract no. DE-AC05-84OR21400 with Martin Marietta Energy Systems, Inc. Publication no. 2918, Environmental Sciences Div., ORNL.

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Received for publication 8 Jan. 1987.

useful half-lives (>1 yr) and ease of counting (both are gamma-emitters).

MATERIALS AND METHODS

Soils

Two contrasting soil profiles were selected for collection of whole soil and macropore materials. Both profiles are located on the Oak Ridge Reservation in Tennessee. One site is at a newly opened sanitary landfill trench in a deep soil profile (Fullerton cherty silt loam) developed on the Knox Group Dolostones. The site was cleared of trees up to 40 yr old in 1979. The Fullerton (Typic Paleudult) is a well-drained, low base-saturated, highly weathered forest soil of the Ultisol order. Soil pH ranges from 4.9 to 5.3 and cation exchange capacity (CEC) ranges from 3 to 10 cmol kg⁻¹. The clay fractions are composed of various amounts of kaolinite, mica, vermiculite, hydroxy-interlayered vermiculite, amorphous iron/aluminum oxide, gibbsite, and quartz (Lee et al. 1984). The second site is in a freshly opened trench across a ravine in a forested area. The soil profile here (Shelocta silt

loam) is shallow and developed on the Maryville Limestone Formation which consists of a limey shale at this location. The Shelocta (Typic Hapludult) is a well-drained, moderately high base-saturated forest soil of the Ultisol order. Soil pH ranges from 4 to 5 and CEC ranges from 10 to 20 cmol kg⁻¹. The clay mineral assemblages are similar to those in the Fullerton, except that micaceous minerals predominate over kaolinite in the Shelocta (Rothschild et al. 1984).

Field methods

Both whole soil and macropore materials (Table 1) were collected at each site. The classification and description of the macropores given below follow those of Beven and Germann (1982). The macropore materials were collected from saprolite-soil contacts (termed *shale clay* or *chert clay*), soil fracture surfaces (termed *fracture clay*), and from tubular-shaped pores formed by plant roots (termed *root clay* or *organic filling*). The soil fractures are probably formed by the shrinking and swelling of the clays. Thin layers, or films, of clay (fracture clay) usually covered the fracture surface. This

TABLE 1
Descriptions of macropore materials

Soil name	Sample identification	Sample type	Depth, m	Thickness of clay, mm	Description
Shelocta	D	Whole soil	2.0	—	Homogenized whole soil from Horizon C
	65	Fracture clay	1.8–2.5	1–3	Brown clay along soil fractures
	66	Root clay	1.5–2.0	2	Gray clay from root channel 4–7 mm in diam.
	69	Shale clay	2.5	2–4	Foliated dark-gray clay covering shale bedrock
	74	Organic filling	1.0	—	Fine, loosely compacted dark organic material from root channels
	79	Root clay	1.0–2.0	5–30	Light-gray clay filling from root channel
Fullerton	A	Whole soil	2.0–3.5	—	Homogenized whole soil from horizon Bt2
	19	Fracture clay	2.0	8	Red clay along soil fracture
	20	Fracture clay	3.0	3	Red clay along soil fracture
	27	Mixed	2.0	—	Mostly red clay and some root bark from a root channel 2–5 mm in diam.
	28	Organic filling	2.0	—	Organic matter associated with sample 27
	31	Chert clay	3.0	3	Red clay mantling a chert clast

coating of clay could be differentiated from the surrounding soil matrix by one or more of the following properties: color, denseness, and shiny luster. The clay film covering rock clasts (shale clay and chert clay) may be a result of translocation of clay from the upper part of the soil profile. The root channels were usually small (<7 mm in diameter) and often contained organic matter. The organic matter (organic filling) often consisted of the remains of root bark that resisted decay longer than the xylem, or consisted of fine, dark, loosely packed organic matter derived primarily from the decaying root. The clay surrounding root channels (root clay) was usually light-gray, compared with the red Fullerton and gray-brown Shelocta soils.

The macropore materials were collected from freshly cut trench faces in the Bt2 horizon at a depth of 2 to 5 m at the Fullerton profile and in the C horizon at a depth of 1.5 to 2.5 m at the Shelocta profile. A small spatula was used to very carefully scrape off an approximately 0.2- to 0.5-mm-thick layer of clay from the surface of a fracture or a weathered rock impression in the soil. Scraping was continued until the appearance of the clay changed; usually 1 to 8 mm of clay was removed. In removing rocks from the soil the macropore material would sometimes adhere to the rock, facilitating its recovery. In the case of root channels, the same method was used, except that an attempt was made to separate the organic filling from the root clay. However, some root bark was inevitably included with the clay scraped from the root channel. One root clay sample (79) was collected from a large (5- to 30-mm diameter) clay-filled root channel at the Shelocta profile.

Approximately 50 kg of whole soil was collected from the Bt2 and C horizons at the Fullerton and Shelocta profiles, respectively. Material coarser than 2 mm was removed by sieving. The <2-mm size fraction was then blended thoroughly and stored in sealed plastic containers to retain natural moisture.

Laboratory methods

Sorptivity was determined as batch distribution coefficients using radiotracers. The distribution coefficient (K_d) is the ratio of concentration of a substance in soil to its concentration in water. Values for K_d were determined on the macropore and homogenized whole soil samples in duplicate using the following solutions: syn-

thetic soil water, pH-modified synthetic soil water, natural soil water, and pH-modified natural soil water. The natural soil water was collected from soil lysimeters near each soil collection site. The ionic composition of the synthetic soil water was based on lysimeter data from the Fullerton soil. The ionic chemical analyses of synthetic and natural soil solutions are presented in Table 2.

The procedure was to place 0.4 g (dry-weight basis, determined on a subsample) of the soil and 4 ml of water into a capped plastic centrifuge tube. Those samples that were designated for pH modification received 0.025 ml of 0.5 *N* H_2SO_4 to ascertain the effect of increased acidity on K_d values. After the tubes were shaken for 16 h on a reciprocating shaker, the samples were centrifuged (500 *gs* for 30 min), and the pH of the supernatant was measured. A stable metal spike (0.050 ml) was then added, followed by 0.050 ml of carrier-free radioisotope. The metal spike solution was designed to yield 1 mg Cd L^{-1} and 10 mg Mn L^{-1} . The radioisotope solution yielded activities for ^{109}Cd and ^{54}Mn of about 1500 cpm ml^{-1} for each isotope after addition to each tube. After an additional 26 h of shaking, the suspensions were again centrifuged and the pH was again measured. With each batch of samples, three tubes without soil standards were processed in the same manner as tubes with soil.

TABLE 2
Chemical composition, $mg L^{-1}$, of solutions used for K_d determinations

	Fullerton Bt2 water	Synthetic water	Shelocta C water
Ca	2.0	1.2	50
Mg	1.3	1.2	7.2
Na	1.5	4.3	4.4
K	1.8	0.64	1.3
Al	0.21	<0.002	0.025
Fe	0.00	<0.005	0.003
Mn	0.29	0.03	0.03
Cd	0.003	5×10^{-5}	1×10^{-4}
Cl	1.6	1.9	2.0
SO_4	14.0	7.9	23.0
NO_3	0.01	0.02	0.15
HCO_3	1.7	8.8	170.0
TOC ^a	4.1	0.9	0.8
pH	5.2	6.6	7.7
EC, $\mu S cm^{-1b}$	44	39	280

^a Total organic carbon.

^b Electrical conductivity.

Two milliliters of supernatant were removed from each centrifuge tube for simultaneous counting of the radioisotopes using a Nuclear Data, Inc., ND6600 Data Acquisition and Processing System and an EG&G ORTEC detector (Model GEM-25200-S with a 55-mm-diameter high-purity germanium crystal). Manganese-54 and cadmium-109 were counted at 835 and 88 keV, respectively, until the counting errors were less than 5% for each isotope.

Elemental content of whole soils and macropore materials was determined by complete digestion in HF-HNO₃, followed by analysis using inductively coupled plasma emission spectrophotometry. Percentage of total carbon was determined by a Leco analyzer.

RESULTS AND DISCUSSION

Because of the very limited quantities of macropore material collected, characterization was limited to the elemental analyses given in Table 3. Clays that lined macropores exhibited elemental compositions similar to those in whole soil from which they were collected. Clay associated with a shale clast in the Shelocta soil was appreciably higher in calcium (Ca) and magnesium (Mg) than the whole soil, as might be expected considering the limey nature of the underlying, less weathered, regolith. Not surprisingly, organic fillings in macropores from both soils were higher in total carbon and lower in most mineral elements than were the whole soils or macropore clays. Incomplete separation of root clay from the organic filling for the root channel sampled in the Fullerton (nos. 27 and 28) is also apparent from the high carbon content (6.59%) of the "mixed" sample. In the subsequent discussion,

macropore materials that contained less than 1% carbon have been termed *mineral*. The high-carbon-content root channel fillings have been termed *organic*.

As shown in Tables 4 and 5, high variability in both solution pH and K_d values was observed between a few pairs of supposedly duplicate samples of macropore materials. For example, one of the root clays from the Shelocta soil exhibited duplicate pH values of 7.4 and 4.7, duplicate K_d values for Cd of 560 and 26, and duplicate K_d values for Mn of 140 and 20. This poor reproducibility is probably attributable to inadvertent acid additions to one of each pair of nonspiked samples.

Although insufficient data were collected to afford a rigorous (statistical) comparison, K_d values at comparable pH values were very similar between synthetic and natural water, especially for the Fullerton samples. The synthetic water composition, especially Ca concentrations, more closely matched the natural water from the Fullerton profile, and thus the better agreement between synthetic and natural water results for the Fullerton samples was not unexpected. Calcium has been shown (Cavallaro and McBride 1978; O'Connor et al. 1984) to compete with Cd for exchange sites on soils. Thus, the absence of a consistent difference between K_d values associated with natural water from the Shelocta (high in Ca, Table 2) and those associated with synthetic water (used with Shelocta material but low in Ca, Table 2) suggests that more than ion exchange is involved in Cd adsorption of these soils and macropore materials.

As illustrated in Figs. 1 through 4, pH is one of the major factors affecting K_d values for whole

TABLE 3
Composition of whole soils and soil macropore linings and fillings; concentrations in percentages

Sample ID:	Shelocta soil						Fullerton soil					
	Whole soil	Fracture clay	Root clay	Root clay	Shale clay	Organic filling	Whole soil	Fracture clay	Fracture clay	Mixed	Organic filling	Chert clay
	(D)	(65)	(66)	(79)	(69)	(74)	(A)	(19)	(20)	(27)	(28)	(31)
C	0.15	0.19	0.38	0.12	0.75	36.5	0.08	0.11	0.60	6.59	38.8	0.13
Ca	0.10	0.05	0.08	0.29	1.13	0.31	0.03	0.01	0.05	0.06	0.02	0.08
Mg	0.63	0.63	0.66	0.59	1.13	0.30	0.47	0.55	0.64	0.48	0.26	0.57
K	3.15	3.12	3.12	2.84	3.79	1.02	1.47	1.68	1.94	1.35	0.55	1.42
Na	0.38	0.37	0.39	0.54	0.53	0.14	0.04	0.05	0.06	0.06	0.03	0.05
Al	7.27	7.14	7.21	6.62	10.2	3.06	8.77	12.4	9.88	8.77	4.36	10.9
Ti	0.43	0.41	0.45	0.41	0.43	0.17	0.31	0.35	0.31	0.29	0.14	0.26
Fe	2.09	2.13	1.37	2.42	3.57	0.68	4.72	5.51	4.83	4.23	1.26	3.05
Mn	0.070	0.074	0.013	0.009	0.017	0.014	0.013	0.013	0.014	0.017	0.008	0.018

TABLE 4

Measured solution pH values (42-h equilibration time) and K_d s for Shelocta whole soil and macropore material

Sample	Replicate	Synthetic water				Natural water			
		No spike		Acid-spiked		No spike		Acid-spiked	
		1	2	1	2	1	2	1	2
Whole soil	pH	4.40	4.50	3.05	3.15	5.10	— ^a	3.55	3.50
	K_d Cd	38	32	3.7	3.4	38	—	6.4	5.6
	K_d Mn	37	32	3.5	4.9	37	—	7.1	6.3
Root clay (66)	pH	4.55	4.55	3.60	3.30	5.20	3.60	3.55	3.65
	K_d Cd	31	30	2.5	2.2	45	3.2	4.2	4.6
	K_d Mn	26	27	2.9	2.5	37	4.2	3.9	4.1
Root clay (79)	pH	6.90	6.60	3.90	3.90	7.40	4.70	4.40	3.40
	K_d Cd	550	490	10	4.7	560	26	23	<0.1
	K_d Mn	180	160	7.7	4.2	140	20	17	1.0
Shale clay (69)	pH	7.2	7.2	6.40	5.60	7.25	6.50	6.30	6.45
	K_d Cd	2050	1200	330	140	1460	280	280	330
	K_d Mn	680	350	120	53	320	89	90	96
Fracture clay (65)	pH	4.20	4.15	3.10	3.10	3.45	4.65	3.20	3.40
	K_d Cd	16	20	3.9	1.9	2.7	32	2.7	3.6
	K_d Mn	16	20	4.7	3.0	4.5	38	3.1	4.3
Organic filling (74)	pH	4.50	4.50	3.95	3.95	4.60	4.65	4.10	4.05
	K_d Cd	480	640	44	105	610	540	130	120
	K_d Mn	140	170	20	31	160	160	29	28

^a Sample was lost.

TABLE 5

Measured solution pH values (42-h equilibration time) and K_d s for Fullerton whole soil and macropore material

Sample	Replicate	Synthetic water				Natural water			
		No spike		Acid-spiked		No spike		Acid-spiked	
		1	2	1	2	1	2	1	2
Whole soil	pH	4.20	4.20	3.45	3.45	4.15	4.10	3.45	3.40
	K_d Cd	12	15	7.5	7.0	20	14	5.6	5.5
	K_d Mn	14	17	9.0	7.8	21	16	6.0	5.7
Mixed (27)	pH	4.65	4.15	3.70	3.80	4.05	4.25	3.70	3.70
	K_d Cd	340	58	13	17	42	74	15	12
	K_d Mn	150	28	6.8	8.9	22	36	6.8	5.4
Fracture clay (19)	pH	3.95	4.10	3.55	3.50	3.90	4.40	3.55	3.40
	K_d Cd	10	20	6.9	5.1	14	57	5.7	4.4
	K_d Mn	12	21	8.0	6.0	15	62	5.7	4.5
Fracture clay (20)	pH	4.20	4.30	3.60	3.65	4.20	4.20	3.55	3.65
	K_d Cd	25	34	10	11	24	24	6.7	11
	K_d Mn	25	31	8	<0.1	23	24	6.7	11
Chert clay (31)	pH	4.45	4.55	3.20	3.85	4.45	4.45	3.95	3.80
	K_d Cd	32	40	12	16	39	38	19	16
	K_d Mn	33	43	13	19	41	41	20	17
Organic filling (28)	pH	4.20	4.20	3.95	3.90	4.05	4.30	3.95	3.70
	K_d Cd	240	280	65	60	110	300	73	28
	K_d Mn	74	73	19	18	38	83	18	8.1

soils and macropore materials. The small additions of acid caused significant decreases in the K_d values for Mn and Cd. The REG procedure of the statistical analysis system (SAS 1982) was used to compare K_d results for whole soils

and macropore materials. For this comparison, any effects of water type (synthetic or natural) were assumed to be negligible. Log K_d values were modeled as dependent on pH. Output from the procedure included the regression coeffi-

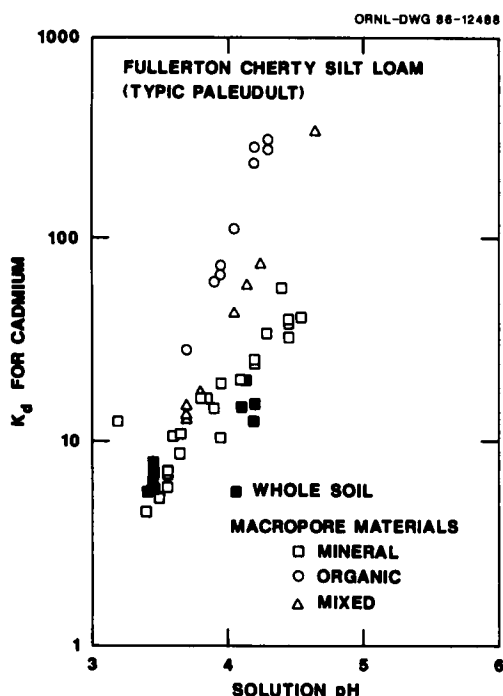


FIG. 1. Effect of pH on the distribution coefficient (K_d) of cadmium on whole soil and various macropore materials from the Bt2 horizon of the Fullerton cherty silt loam.

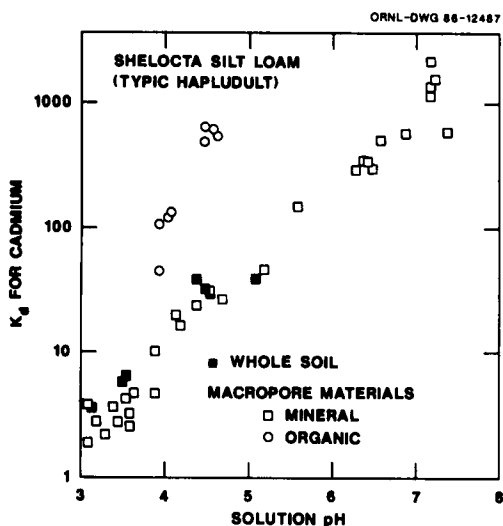


FIG. 2. Effect of pH on the distribution coefficient (K_d) of cadmium on whole soil and various macropore materials from the C horizon of the Shelocta silt loam.

cients and statistics given in Table 6. The high r^2 values attest to the high dependence of K_d values on pH. Likewise, the effect of organic matter on K_d values is reflected in the higher

regression slopes for the samples with high total carbon content.

Various combinations of the sample types were tested statistically for common slope of the log K_d /pH relationship by comparing individual

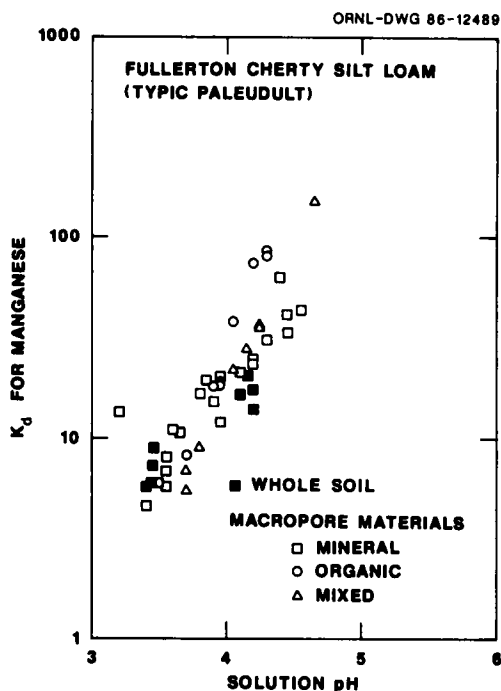


FIG. 3. Effect of pH on the distribution coefficient (K_d) of manganese on whole soil and various macropore materials from the Bt2 horizon of the Fullerton cherty silt loam.

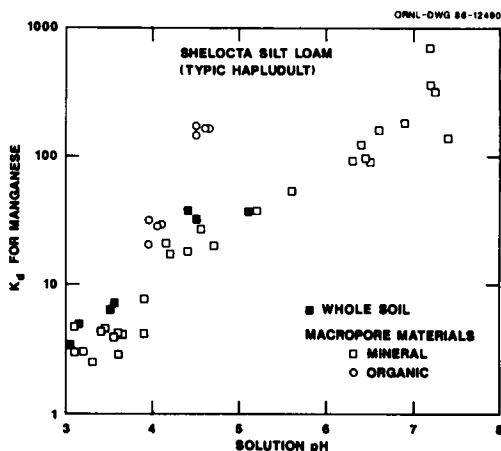


FIG. 4. Effect of pH on the distribution coefficient (K_d) of manganese on whole soil and various macropore materials from the C horizon of the Shelocta silt loam.

TABLE 6

Summary of regression coefficients and statistics for relationships between $\log K_d$ values and solution pH

Element	Soil	Sample type	Slope	Standard error	r^2
Cd	Shelocta	Whole	0.60	0.07	0.92
		Fracture	0.80	0.13	0.88
		Organic	1.45	0.17	0.91
		Clast	0.69	0.08	0.92
		Root	0.63	0.04	0.96
	Fullerton	Whole	0.50	0.07	0.87
		Fracture	0.94	0.07	0.93
		Organic	1.83	0.11	0.98
		Clast	0.42	0.05	0.92
		Mixed	1.45	0.04	0.99
Mn	Shelocta	Whole	0.56	0.07	0.92
		Fracture	0.70	0.09	0.92
		Organic	1.32	0.13	0.94
		Clast	0.61	0.09	0.86
		Root	0.47	0.04	0.92
	Fullerton	Whole	0.51	0.07	0.87
		Fracture	0.91	0.07	0.92
		Organic	1.81	0.12	0.97
		Clast	0.41	0.05	0.92
		Mixed	1.43	0.04	0.99

and pooled regression residuals using the F distribution. For the Shelocta silt loam this analysis showed no significant differences (1% level) among the whole soil and the mineral-type macropore materials (fracture, clast, and root clays) in the slopes of the $\log K_d$ /pH relationships for either cadmium or manganese. The slopes for the organic macropore fillings were significantly different (1% level) from the essentially "mineral" whole soil and the macropore clays. For the Fullerton cherty silt loam the comparison of regression slopes also revealed no significant differences among whole soil and the mineral-type macropore materials. The slopes for the $\log K_d$ /pH relationships for organic macropore fillings and the mixed clay/organic matter material were not significantly different (1% level), but were significantly different (1% level) from the slopes for the whole soil and macropore clays.

The relative importance of soil organic matter versus soil clays as sorbents for Cd and Mn is well illustrated by these results. The organic macropore fillings exhibited higher sorptivities for Cd and Mn than did the whole soils or macropore clays, and the sorptivities increased with increasing pH faster for the organic fillings than for the whole soils or macropore clays. The differences we observed in regression slopes ($\log K_d$ versus pH) for mineral and organic materials

are very similar to those noted by Helling et al. (1964) for the relationships between CEC and pH for clay and organic matter in 60 Wisconsin soils. In both cases the higher density of pH-dependent sorption sites on soil organic matter results in steeper regression slopes for soil organic matter than for clays.

CONCLUSIONS

Results of this study indicated that organic-rich linings and fillings in the root-derived soil macropores exhibit higher sorptivities for Cd and Mn than does whole soil, particularly at lower solution pH values (pH 4 to 5). Due to the higher density of pH-dependent sorption sites on organic matter, sorption on root-derived macropore materials increases more rapidly as pH increases than does sorption on inorganic macropore materials. Inorganic linings and fillings from macropores exhibited sorptivities for Cd and Mn that were very similar to whole soil sorptivities. For both organic and inorganic macropore materials, solution pH was generally far more important in determining sorptivity than the nature of the macropore material. Overall, the results of the study suggest that the migration of Cd and Mn will likely be more highly retarded in root-derived macropores than in other types of macropores. The absence of significant differences in sorptivity between inorganic macropore materials and whole soil suggests that factors other than sorptivity (e.g., hydrology) may be more important in determining whether retardation of Cd and Mn is greater or less in the whole soil matrix than in non-root-derived macropores.

REFERENCES

- Baes, C. F., III, and R. D. Sharp. 1983. A proposal for estimation of soil leaching and leaching constants for use in assessment models. *J. Environ. Qual.* 12(1):17-28.
- Beven, K., and P. Germann. 1982. Macropores and water flow in soils. *Water Resour. Res.* 18(5):1311-1325.
- Cavallaro, N., and M. B. McBride. 1978. Copper and cadmium adsorption characteristics of selected acid and calcareous soils. *Soil Sci. Soc. Am. J.* 42:550-556.
- Helling, C. S., G. Chesters, and R. B. Corey. 1964. Contribution of organic matter and clay to soil cation exchange capacity as affected by the pH of the saturating solution. *Soil Sci. Soc. Am. Proc.* 28:517-520.
- Lee, S. Y., O. C. Kopp, and D. A. Lietzke. 1984. Mineralogical characterization of west Chestnut Ridge soils. ORNL/TM-9361. Oak Ridge National Laboratory, Oak Ridge, Tenn.

- O'Connor, G. A., C. O'Connor, and G. R. Cline. 1984. Sorption of cadmium by calcareous soils: Influence of solution composition. *Soil Sci. Soc. Am. J.* 48:1244-1247.
- Rothschild, E. R., D. D. Huff, B. P. Spalding, S. Y. Lee, R. B. Clapp, D. A. Lietzke, R. G. Stansfield, N. D. Farrow, C. D. Farmer, and I. L. Munro. 1984. Characterization of soils at proposed solid waste storage area (SWSA) 7. ORNL/TM-9326. Oak Ridge National Laboratory, Oak Ridge, Tenn.
- Shuford, J. W., D. D. Fritton, and D. E. Baker. 1977. Nitrate-nitrogen and chloride movement through undisturbed field soil. *J. Environ. Qual.* 6(3):255-259.
- Statistical Analysis System (SAS). 1982. SAS user's guide: Statistics. SAS Institute, Cary, N.C.
- Thomas, G. W., and R. E. Phillips. 1979. Consequences of water movement in macropores. *J. Environ. Qual.* 8(2):149-152.